

Module 09: Sliding Window Protocols:

Selective Repeat ARQ Architecture

Unit 2: Data Link Layer & MAC Sublayer

Out-of-Order Buffering & NAK Mechanics

In-depth architectural analysis of Selective Repeat ARQ, receiver reassembly buffer logic, NAK selective rejection, and formal mathematical derivation of window sizing $W_S = W_R = 2^{m-1}$.

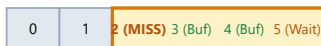
Selective Repeat ARQ: Independent Buffering, NAK Retransmission & Window Sizing Proof

1. Window Architecture ($m = 3 \text{ bits} \rightarrow W_S = W_R = 4$)

Sender Window ($W_S = 2^{m-1} = 4$):



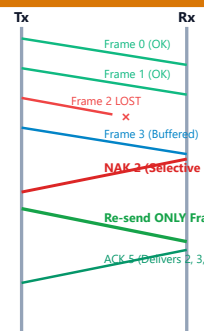
Receiver Window ($W_R = 4$, Has Out-of-Order Buffer):



• Key Architectural Advantage:

Frames 3 & 4 are STORED in Receiver Buffer, NOT discarded!
Sender ONLY retransmits the missing Frame 2 (Zero wastel).

2. Timeline: NAK 2 & Selective Retransmission



• Selective Repeat Merits:

1. Receiver buffers out-of-order frames; no unnecessary discards.
2. Immediate NAK triggers quick retransmission before timer expiry.

• Trade-off / Complexity:

Requires independent timer per frame and sorting reassembly buffers at the receiver side.

Mathematical Proof: Why $W_S = W_R = 2^{m-1}$? (AKTU Core Exam Derivation)

Proof ($W_S + W_R \leq 2^m$):

For no overlap between old and new sequence numbers: $\text{Window_Sender} + \text{Window_Receiver} \leq \text{Total Sequences} = 2^m$.

Since receiver must buffer up to W_S frames to avoid buffer overflow, we choose $W_S = W_R$. Therefore: $2 \cdot W_S \leq 2^m \rightarrow W_S \leq 2^{m-1}$.

Counter-Example ($W_S > 2^{m-1}$):

If $m=2$ (sequences 0,1,2,3) and $W_S=W_R=3$: Sender transmits 0,1,2. Receiver accepts all and advances window to {3, 0, 1}.

If all ACKs are lost, sender retransmits old Frame 0. Receiver window contains 0 (as NEW frame)! Overlap disaster occurs!

Master Rule:

Stop-and-Wait: $W_S=1, W_R=1$ | **Go-Back-N:** $W_S=2^m - 1, W_R=1$ | **Selective Repeat:** $W_S = W_R = 2^{m-1}$

1. Selective Repeat vs Go-Back-N

While Go-Back-N discards all subsequent frames upon encountering a gap, Selective Repeat accepts and buffers out-of-order frames ($W_R = 2^{m-1}$). It selectively requests retransmission of only the missing frame via NAK frames.

2. Formal Proof of $W_S = W_R = 2^{m-1}$

Window Boundary Constraint

$$W_{\text{Sender}} + W_{\text{Receiver}} \leq 2^m$$

$$\text{Since } W_S = W_R: 2 \cdot W_S \leq 2^m \rightarrow W_S \leq 2^{(m-1)}$$

If $W_S > 2^{(m-1)}$, duplicate frames from an unacknowledged old window will map directly into the advance receiver window, producing undetectable data corruption.